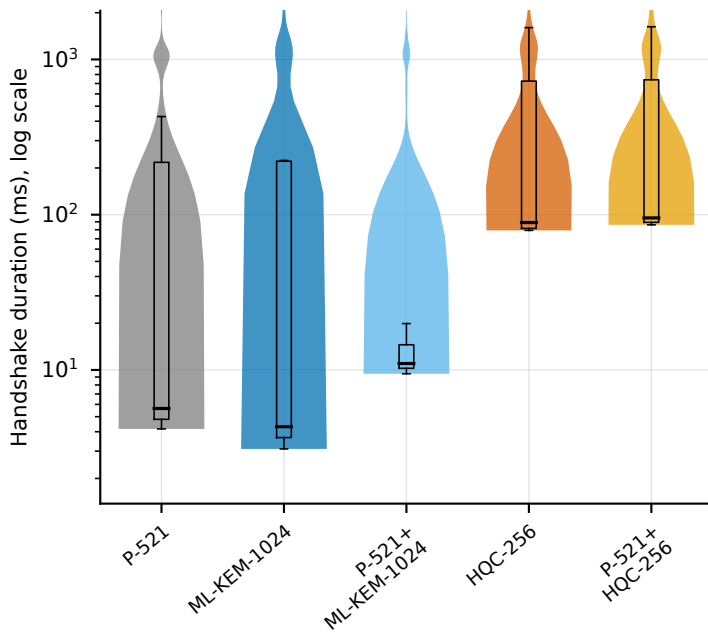
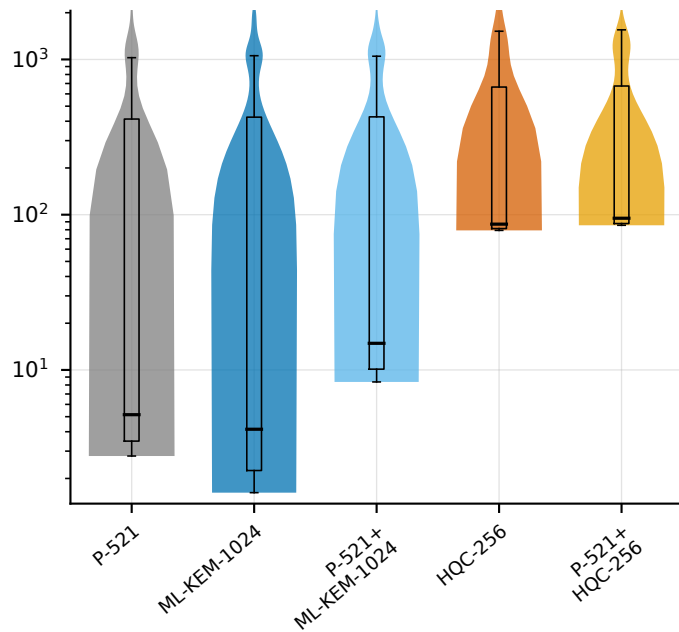


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	258.8	206.2	4.30k	363.0	212.7	9.55k
ML-KEM-1024	388.3	244.3	13.23k	398.6	168.0	4.18k
P-521+ML-KEM-1024	184.3	239.1	3.16k	406.5	183.7	6.60k
HQC-256	478.4	148.9	7.32k	564.9	215.2	14.02k
P-521+HQC-256	503.9	150.4	7.59k	532.9	153.6	6.44k